INCH-POUND
MIL-C-29600/24 (AS)
22 February 1990

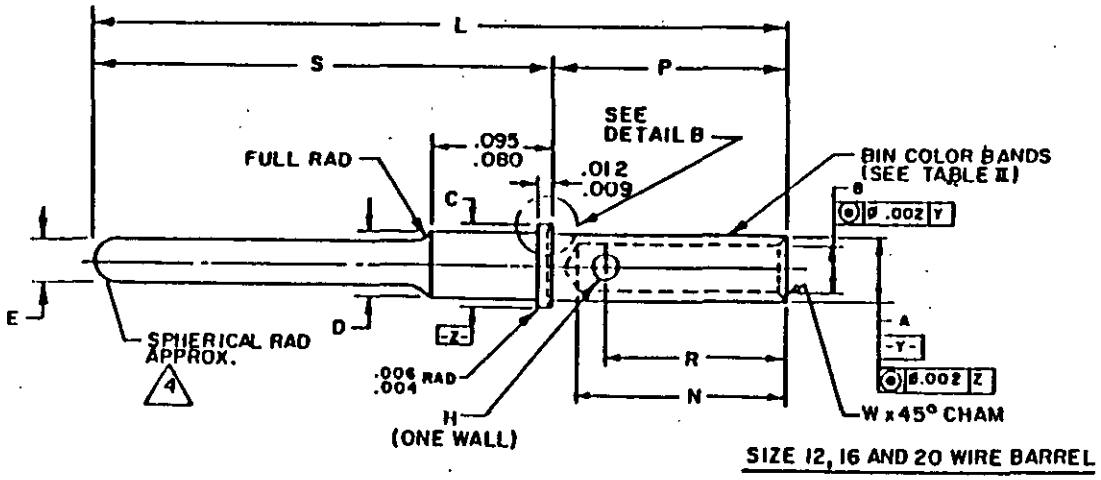
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, MINIATURE, COMPOSITE, HIGH DENSITY, QUICK COUPLING, ENVIRONMENT RESISTANT, CONTACTS, PIN, CRIMP REMOVABLE, (SERIES B)

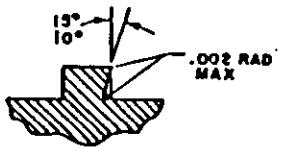
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest revision of MIL-C-29600.

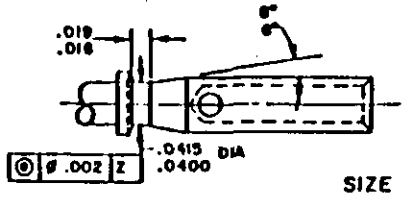
INACTIVE FOR NEW DESIGN



SIZE 12, 16 AND 20 WIRE BARREL



DETAIL B



SIZE 22 WIRE BARREL

NOTES:

1. Dimensions are in inches unless otherwise specified.
2. Metric equivalents are based upon 1 inch = 25.4 mm.
3. Dimensions apply after plating.
4. Spherical end dimension shall conform to MIL-C-39029.
5. Manufacturer's trademark per AIR-1351

AMSC N/A FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T00:41Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions.

BIN Code	A	B	C	D	E	H	L Ref	N	P	R	S	W Cham
610	.0480	.0355	.0555	.0390	.0275	.022	.636	.155	.218	.145	.426	.006
	.0465	.0335	.0540	.0365	.0260	.018		.145	.208	.139	.420	.003
611	.0615	.0450	.0760	.0615	.0410	.027	.636	.195	.218	.165	.426	.007
	.0600	.0430	.0745	.0600	.0390	.023		.185	.208	.159	.420	.004
612	.0925	.0645	.1080	.0925	.0630	.027	.636	.195	.218	.165	.426	.010
	.0910	.0625	.1065	.0910	.0610	.023		.185	.208	.159	.420	.006
613	.1515	.1020	.1695	.1515	.0950	.042	.663	.232	.245	.196	.426	.012
	.1500	.0980	.1680	.1500	.0930	.038		.216	.235	.190	.420	.006

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.0030	.0762	.0380	.9652	.0760	1.930	.1680	4.267
.0040	.1016	.0390	.9906	.0910	2.311	.1695	4.305
.0060	.1524	.0410	1.041	.0925	2.350	.1850	4.699
.0070	.1778	.0420	1.067	.0930	2.362	.1900	4.826
.0100	.2540	.0430	1.092	.0950	2.413	.1950	4.953
.0120	.3038	.0450	1.143	.0980	2.489	.1960	4.978
.0150	.3810	.0465	1.181	.1020	2.591	.2080	5.283
.0180	.4572	.0480	1.219	.1065	2.705	.2160	5.486
.0220	.5588	.0540	1.372	.1080	2.743	.2180	5.537
.0230	.5842	.0555	1.410	.1390	3.531	.2320	5.893
.0260	.6604	.0600	1.524	.1450	3.683	.2350	5.969
.0270	.6858	.0610	1.549	.1500	3.810	.2450	6.223
.0275	.6985	.0615	1.562	.1515	3.848	.4200	10.67
.0335	.8509	.0625	1.588	.1550	3.937	.4260	10.82
.0355	.9017	.0630	1.600	.1590	4.039	.6360	16.15
.0365	.9271	.0645	1.638	.1650	4.191	.6630	16.84

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:41Z
Check the source to verify that this is the current version before use.

TABLE II. Design characteristics.

BIN code	Color bands			Mating end size	Wire barrel size
	1st	2nd	3rd		
610	Blue	Brown	Black	23	22
611	Blue	Brown	Brown	20	20
612	Blue	Brown	Red	16	16
613	Blue	Brown	Orange	12	12

TABLE III. Tools.

Wire barrel size	Basic crimping tool	Positioner	Installing tool	Removal tool	
				Wired	Unwired
22	M22520/2-01	M22520/2-13	M81969/16-04	M81969/16-04	M81969/30-01
20	M22520/2-01	M22520/2-14	M81969/16-01	M81969/16-01	M81969/30-02
16	M22520/2-01	M22520/2-15	M81969/16-02	M81969/16-02	M81969/30-03
12	M22520/1-01	M22520/1-09	M81969/16-03	M81969/16-03	M81969/30-04

REQUIREMENTS:

Dimensions: See Figure and Table I.

Design characteristics and configuration: See Figure and Table II.

Tools: See Table III.

Mating contacts: MIL-C-29600/23.

Temperature: -65°C to +175°C.

Durability: 1500 cycles.

Vibration: The vibration requirements shall consist of those tests specified in MIL-C-29600. The contact sample group shall be 12 contacts, divided equally between the three separate vibration test of MIL-C-29600.

Shock: The shock requirements shall consist of those tests specified in MIL-C-29600. The contact sample group shall be divided in half and each group subjected to a separate shock test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:41Z
Check the source to verify that this is the current version before use.

REQUIREMENTS (continued)

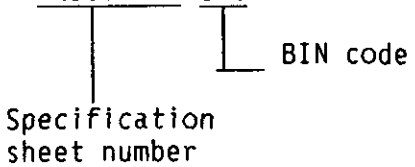
Qualification: The qualification requirements shall consist of the sample numbers and tests specified in MIL-C-39029 for type A contacts with localized finish. The vibration test group shall consist of double the number of samples specified in MIL-C-39029. For those tests which require the use of a connector pair, either a composite or metal shell connector pair may be utilized. Caution should be used if exceeding the rated durability cycles of a metal shell connector. For vibration and shock testing, accessories shall be as specified in MIL-C-29600.

Periodic qualification: The manufacturer shall forward a report every 24 months to the qualifying activity. The periodic qualification requirements shall consist of those tests specified for qualification except for vibration and shock.

Quality conformance inspection: The quality conformance requirements shall consist of Groups A and B inspection as specified in MIL-C-39029.

Material: Base: Conductive copper alloy.
Plating: See 3.3.6 of MIL-C-29600.

Example of part number: M29600/24-610



Custodians:
Navy - AS

Preparing activity:
Navy - AS
(Project No. 5935-3701-08)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T00:41Z
Check the source to verify that this is the current version before use.